Boolean Networks

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Boolean Abstraction

Boolean networks constitute a "top-down" abstraction, allowing us to model systems which are not understood in full detail.

In general, Boolean networks are well suited for modelling systems with the following properties:

- Made up of interacting entities.
- Each entity is characterised by a variable quantity.
- The events (and their mechanisms) are not directly observable, only their complete consequences.

Boolean Networks – Syntax

A Boolean network f of dimension $n \in \mathbb{N}$ (on n variables) is a collection of n-ary Boolean functions $f_1, \ldots, f_n$:

$$f = (f_1, \ldots, f_n)$$

For each $i \in \{1, \ldots, n\}$, $f_i : \mathbb{B}^n \to \mathbb{B}$ is the **local function** of the i-th variable.

The Boolean network itself can be understood as a Boolean vector function $f : \mathbb{B}^n \to \mathbb{B}^n$, defined as $f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_n(\mathbf{x}))$.

EXAMPLE:

$x : \{1, \ldots, n\} \to \mathbb{B}$ "valuation" $\Rightarrow$ equivalent
$x \in \mathbb{B}^n$ Boolean vector of length n

$$f_1(\mathbf{x}) = \mathbf{x}_3 \wedge (\neg \mathbf{x}_1 \vee \neg \mathbf{x}_2)$$
$$f_2(\mathbf{x}) = \mathbf{x}_1 \wedge \mathbf{x}_3$$
$$f_3(\mathbf{x}) = \mathbf{x}_1 \vee \mathbf{x}_2 \vee \mathbf{x}_3$$

Boolean Networks – Semantics

A Boolean network defines a finite dynamical system with discrete states (**transition system**). A transition system $(S, \rightarrow)$ is composed of:

- A finite set of states S;
- An irreflexive transition relation $\rightarrow \subseteq S \times S$;
  ↪ "choice"

We use the natural infix notation for the transition relation, $\mathbf{x} \rightarrow \mathbf{y}$ instead of $(\mathbf{x}, \mathbf{y}) \in \rightarrow$.

For a Boolean network of dimension n, the set of states is the set of all Boolean vectors of length n, $S = \mathbb{B}^n$.

EXAMPLE: $(0,0,0) = 000$
For dimension $n = 3$, $S = \{000, 001, 010, 100, 011, 101, 110, 111\}$.

The exact shape of the transition relation is determined by the chosen **update mode**.
Multiple different update modes are commonly employed.

Update Modes

Given a configuration $\mathbf{x} \in \mathbb{B}^n$, then for each variable $i \in \{1, \ldots, n\}$, $f_i(\mathbf{x})$ is the "next value" of the i-th variable.

In particular a variable such that $\mathbf{x}_i \neq f_i(\mathbf{x})$ is called **frustrated** (in the configuration $\mathbf{x}$).

There are multiple ways of upgrading configurations:

- All variables update value simultaneously;
- Variables update value sequentially;
- Variables are split into blocks and each block updates values simultaneously;
- ... each block updates values sequentially;
- Internal clocks;
- ...

Update Functions

Update functions are a family of functions which are restating the local functions $f_1, \ldots, f_n$ as direct modification of the configurations, $\Phi : \mathbb{B}^n \to \mathbb{B}^n$.

For any subset of variables $W \subseteq \{1, \ldots, n\}$, the update function $\Phi_W : \mathbb{B}^n \to \mathbb{B}^n$ is defined as $\Phi_W : \mathbf{x} \mapsto \mathbf{y}$ where for each $i \in \{1, \ldots, n\}$:

$$\mathbf{y}_i \triangleq \begin{cases} f_i(\mathbf{x}) & \text{if } i \in W \\ \mathbf{x}_i & \text{if } i \notin W \end{cases}$$

Examples:

$\Phi_\emptyset = id : \mathbf{x} \mapsto \mathbf{x}$

$\Phi_{\{1,\ldots,n\}} : \mathbf{x} \mapsto (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_n(\mathbf{x}))$

$\forall i \in \{1, \ldots, n\}, \Phi_{\{i\}} = \Phi_i : \mathbf{x} \mapsto (\mathbf{x}_1, \ldots, \mathbf{x}_{i-1}, f_i(\mathbf{x}), \mathbf{x}_{i+1}, \ldots, \mathbf{x}_n)$

Transitions defined by means of update functions, $\mathbf{x} \to \Phi_{W_p} \circ \cdots \circ \Phi_{W_1}(\mathbf{x})$ for some $W_1, \ldots W_p \subseteq \{1, \ldots, n\}$, are called **elementary**.

Synchronous Semantics

"All variables have their value updated simultaneously."

Synchronous semantics only uses one update function, $\Phi_{\{1,\ldots,n\}} : \mathbf{x} \mapsto (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_n(\mathbf{x}))$. Formally:

$$\forall \mathbf{x} \neq \mathbf{y} \in \mathbb{B}^n, \mathbf{x} \xrightarrow{sync} \mathbf{y} \overset{\Delta}{\Longleftrightarrow} \mathbf{y} = \Phi_{\{1,\ldots,n\}}(\mathbf{x})$$

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Deterministic – Each configuration has at most one successor.

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$$100 \xrightarrow{sync} 001 \qquad 001 \xrightarrow{sync} 101$$
$$010 \xrightarrow{sync} 001 \qquad 101 \xrightarrow{sync} 111$$
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Fully Asynchronous Semantics

"Only one variable updates value in each transition."

Fully asynchronous semantics use all singleton update functions, $\Phi_i : \mathbf{x} \mapsto (\mathbf{x}_1, \ldots \mathbf{x}_{i-1}, f_i(\mathbf{x}), \mathbf{x}_{i+1}, \ldots, \mathbf{x}_n)$, $i \in \{1, \ldots, n\}$. Formally:

$$\forall \mathbf{x} \neq \mathbf{y} \in \mathbb{B}^n, \mathbf{x} \xrightarrow{async} \mathbf{y} \overset{\Delta}{\Longleftrightarrow} \exists i \in \{1, \ldots, n\}, \mathbf{y} = \Phi_i(\mathbf{x})$$

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Generalised Asynchronous Semantics

"Any subset of variables has their values updated simultaneously."

Generalised asynchronous semantics use all update functions, Φ_W for $W \subseteq \{1, \ldots, n\}$. Formally:

$$\forall \mathbf{x} \neq \mathbf{y} \in \mathbb{B}^n, \mathbf{x} \xrightarrow{gen} \mathbf{y} \overset{\Delta}{\Longleftrightarrow} \exists W \subseteq \{1, \ldots, n\}, \mathbf{y} = \Phi_W(\mathbf{x})$$

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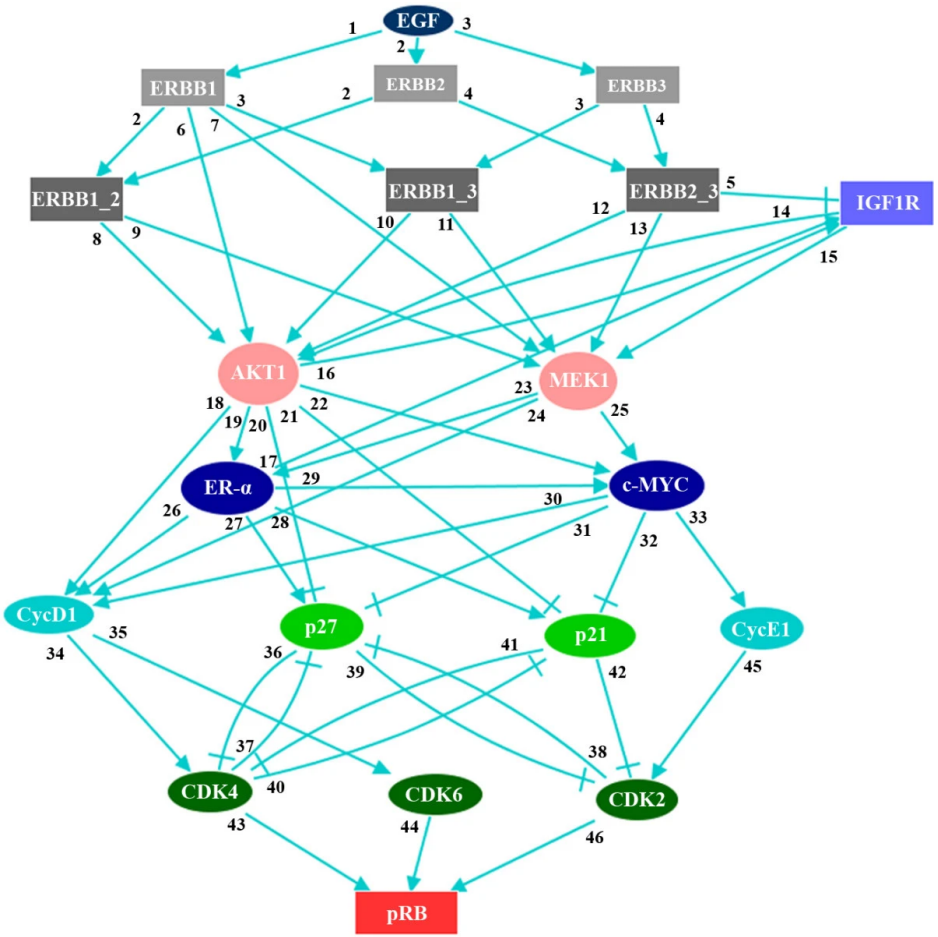
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Example: Drug Target Discovery



Target	Logical rules for the activation of target
ERBB1	$f_{ERBB1}(x)$=EGF
ERBB2	EGF
ERBB3	EGF
ERBB1_2	ERBB1 ∧ ERBB2
ERBB1_3	ERBB1 ∧ ERBB3
ERBB2_3	ERBB2 ∧ ERBB3
IGF1R	(ER-α ∨ AKT1) ∧ ! ERBB2_3
ER-α	AKT1 ∨ MEK1
c-MYC	AKT1 ∨ MEK1 ∨ ER-α
AKT1	ERBB1 ∨ ERBB1_2 ∨ ERBB1_3 ∨ ERBB2_3 ∨ IGF1R
MEK1	ERBB1 ∨ ERBB1_2 ∨ ERBB1_3 ∨ ERBB2_3 ∨ IGF1R
CDK2	Cyclin E1 ∧ ! p21 ∧ ! p27
CDK4	Cyclin D1 ∧ ! p21 ∧ ! p27
CDK6	Cyclin D1
Cyclin D1	AKT1 ∨ MEK1 ∨ ER-α ∨ c-MYC
Cyclin D1*	ER-α ∧ c-MYC ∧ (AKT1 ∨ MEK1)
Cyclin E1	c-MYC
p21	ER-α ∧ ! AKT1 ∧ !c-MYC ∧ ! CDK4
p27	ER-α ∧ ! CDK4 ∧ ! CDK2 ∧ ! AKT1 ∧ !c-MYC
pRB	(CDK4 ∧ CDK6) ∨ (CDK4 ∧ CDK6 ∧ CDK2)

Ö. Sahin, H. Fröhlich, C. Löbke, U. Korf, S. Burmester, M. Majety, J. Mattern, I. Schupp, C. Chaouiya, D. Thieffry, A. Poustka, S. Wiemann, T. Beissbarth, and D. Arlt. Modeling erbb receptor-regulated g1/s transition to find novel targets for de novo trastuzumab resistance.

BMC Systems Biology, 3(1):1, Jan 2009

Multivalued Extensions

A maximum value for each variable, $m_1, \ldots, m_n \geq 1$, giving us the following state space $S = \{0, \ldots, m_1\} \times \{0, \ldots, m_2\} \times \cdots \times \{0, \ldots, m_n\}$.

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THOMAS NETWORKS (Discrete multivalued networks with thresholds)

The network definition remains unchanged $f = (f_1, \ldots, f_n) \colon \mathbb{B}^n \to \mathbb{B}^n$, but additionally an $n \times n$ matrix T of thresholds has to be specified.

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Where:
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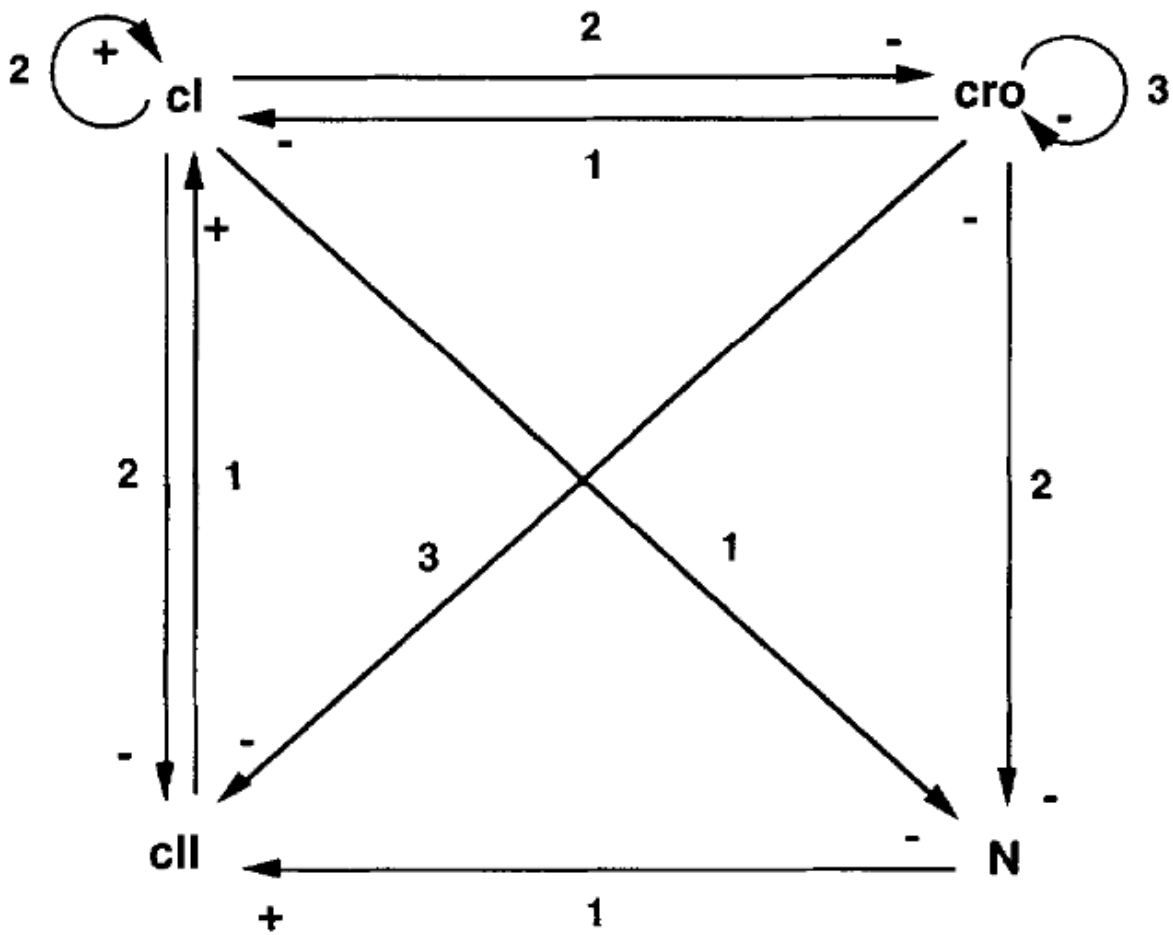
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$\Phi_W \colon S \to S$
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$$\mathbf{y}_i \stackrel{\Delta}{=} \begin{cases} max(0, \mathbf{x}_i - 1) & \text{if } i \in W \\ & \text{and } f_i(\mathbf{x}) = 0 \\ min(m_i, \mathbf{x}_i + 1) & \text{if } i \in W \\ & \text{and } f_i(\mathbf{x}) = 1 \\ \mathbf{x}_i & \text{if } i \notin W \end{cases}$$

Example: Bacteriophage Lambda



D. Thieffry and R. Thomas. Dynamical behaviour of biological regulatory networks—ii. immunity control in bacteriophage lambda.

Bulletin of Mathematical Biology, 57(2):277–297, 1995